

①

	Annual revenue	Net Profit
Acme Computer	138	8
Nadir Software	87	5
Quixote Media	111	13

a) Quixote had largest annual revenue

Ⓕ

b) Nadir had lowest net profit and Acme Computer had largest annual revenue

Ⓓ

c) Acme Computer had the largest net profit or Quixote had the largest net profit

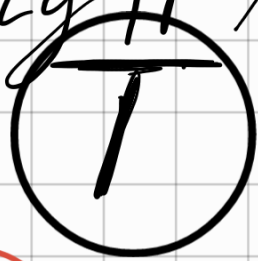
Ⓓ

d) If Quixote had the smallest net profit then Acme had the largest annual revenue

Ⓓ

-?

e) Nadir had the smallest net Profit if and only if Acme had the largest annual rev



② P — It is below freezing

q — It is snowing

a) It is below freezing and snowing.

$P \wedge q$ (P and q)

b) It is below freezing but not snowing

$P \wedge \neg q$

c) It is not below freezing and it is not snowing

$\neg P \wedge \neg q$

d) It is either snowing or below freezing

$q \vee P$



e) If it is below freezing it is also snowing
 $P \rightarrow q$

f) Either it is below freezing or it is snowing
but it is not snowing if it is below freezing
 $P \oplus q$ -? $(P \vee q) \wedge (P \rightarrow \neg q)$

g) That it is below freezing is necessary
and sufficient for it to be snowing
 $q \leftrightarrow P$ -? $P \leftrightarrow q$

③ a) $P \rightarrow \neg P$ 2^n $2^1 = 2$

b) $(P \vee \neg r) \wedge (q \vee \neg s)$ $2^4 = 16$

c) $q \vee P \vee \neg s \vee \neg r \vee \neg t \vee u$ $2^6 = 64$

d) $(P \wedge r \wedge t) \leftrightarrow (q \wedge t)$ $2^4 = 16$

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a) $p \wedge \neg p$

b) $p \vee \neg p$

p	$\neg p$
T	F
F	T

$p \wedge \neg p$
F
F

$p \vee \neg p$
T
T

c) $(p \vee \neg q) \rightarrow q$

d) $(p \vee q) \rightarrow (p \wedge q)$

p	q	$\neg q$	$(p \vee \neg q) \rightarrow q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	F	T	T
T	F	T	F	F
F	T	F	T	F
F	F	T	F	T?

$$e) (p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$$

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$	$(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

$$f) (p \rightarrow q) \rightarrow (q \rightarrow p)$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \rightarrow (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

5 Answer: 2 Professors wanted coffee.
Question was "Does everyone want coffee?"
1st P. says "Idk" b.c. he wants but not
sure if other two want
2nd P. says "Idk" - Same as 1st
3rd P. says "No, not everyone wants"
means that he doesn't want and
first two does.

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a) Jan is rich and happy
- Jan is not rich or Jan is not happy
b) Carlos will bicycle or run tomorrow
- Carlos will not bicycle and will not
run tomorrow

c) Mei walks or takes the bus to class

- Mei doesn't walk and doesn't take bus

d) Ibrahim is smart and hard working

- Ibrahim isn't smart or isn't hard working

7 a) $(P \wedge Q) \rightarrow P$

P	Q	$P \wedge Q$	$(P \wedge Q) \rightarrow P$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

$$b) P \rightarrow (P \vee q)$$

P	q	$P \vee q$	$P \rightarrow (P \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

$$c) \neg P \rightarrow (P \rightarrow q)$$

P	q	$\neg P$	$P \rightarrow q$	$\neg P \rightarrow (P \rightarrow q)$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

$$d) (p \wedge q) \rightarrow (p \rightarrow q)$$

p	q	$p \wedge q$	$p \rightarrow q$	$(p \wedge q) \rightarrow (p \rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

$$e) \neg(p \rightarrow q) \rightarrow p$$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg(p \rightarrow q) \rightarrow p$
T	T	T	F	T
T	F	F	T	T
F	T	T	F	T
F	F	T	F	T

$$f) \neg(P \rightarrow q) \rightarrow \neg q$$

$\neg q$	$\neg(P \rightarrow q) \rightarrow \neg q$
F	T
T	T
F	T
T	T

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a) $\exists x (P(x) \wedge Q(x))$

b) $\exists x (P(x) \wedge \neg Q(x))$

c) $\forall x (P(x) \vee Q(x))$

d) $\forall x (\neg P(x) \vee \neg Q(x))$

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a) $\forall n (n^2 \geq 0) - T$

b) $\exists n (n^2 = 2) - F$

c) $\forall n (n^2 \geq n) - T$

d) $\exists n (n^2 < 0) - F$

12 $P(x) \ x = \{0, 1, 2, 3, 4\}$

a) $\exists x P(x)$

$$P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$$

b) $\forall x P(x)$

$$P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$$

c) $\exists x \neg P(x)$

$$\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$$

$$d) \forall x \neg P(x)$$

$$\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$$

$$e) \neg \exists x P(x)$$

$$\neg (P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$$

$$f) \neg \forall x P(x)$$

$$\neg (P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$$